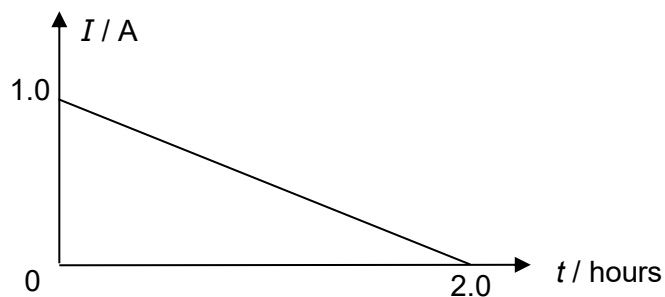


- 19 A portable fan battery is charged by connecting it to a constant potential difference of 6.0 V. The variation with time t of the current I through the battery is as shown.



What is the energy transferred to the battery during the time of 2.0 hours shown in the graph?

- A** 360 J **B** 720 J **C** 22 000 J **D** 43 000 J

Ans: C

$$Q = \text{Area under graph} = \frac{1}{2}(1.0)(2.0 \times 3600) = 3600 \text{ C}$$

$$\text{Energy transferred i.e. work done on charge} = QV = 3600(6.0) = 21600 \text{ J}$$

20 The $I-V$ characteristics of two electrical components P and Q are shown below